11 February 1957

The Honourable J. J. Cahill MLA, Premier of New South Wales
for the Government of New South Wales, Premier's Department, Sydney NSW

CONFIDENTIAL

Dear Mr Cahill,

Re: Fee proposal to provide Architectural Services
Project: National Opera House, Bennelong Point, Sydney

It was a great honour to learn that our design was awarded first prize in your Government's international competition, announced on 29 January 1957, and we thank you for the opportunity to discuss the next steps for the project.

Further to the announcement and our subsequent correspondence, we include here:
- a summary of our assessment of the design brief and site assessment;
- an initial forecast of the Cost of Works for the project, and
- the method for calculating our fee for our services as your architect,

all for your consideration.

A draft Client Architect Agreement is attached and provides detailed information on the services to be provided and contractual conditions of our engagement as your Architect. This agreement is the Australian Institute of Architects' standard-form Client Architect Agreement 2024 (the CAA). We also attach the 'Information guide to the client' that helps you understand the CAA, as well as a copy of the applicable Code of Conduct required under the Architects Act that applies to all architects (for a site in New South Wales, the code of conduct under the Architects Act 2003 (NSW)).

Design Brief and Site Assessment

We have summarised our understanding of your preliminary design brief in accordance with our recent discussions to allow us to evaluate the approximate size and complexity of the project.

Our understanding of the brief is taken from the competition's Conditions and Programme (1955) and from our winning scheme. The project includes:
- a large hall seating about 3,000 people, for symphony concerts, opera, ballet, choral concerts and large public meetings;
- a small hall seating about 1,200 people, for drama, smaller-scale opera, chamber music, recitals and lectures;
- rehearsal rooms, a restaurant, meeting rooms, and the stage, dressing and back-of-house areas both halls need;
- a raised platform (the podium) holding the stage machinery, building services and the public approach and foyers; and
- the roof: a series of shell vaults rising over both halls.

Site: Bennelong Point, Sydney Harbour (at present the Fort Macquarie tram depot).

Site assessment: the site is prominent and surrounded by water on three sides. We have not yet seen a survey of the ground conditions. Because the podium and its columns will carry the roof, the ground must be investigated before the foundations are designed, so we recommend a geotechnical investigation early in the Concept Design stage (see Specialist Consultants below).

Note: The seat numbers and uses outlined above are initial, indicative only and are likely to change as the design brief and project design evolve. Floor areas will be confirmed at the Concept Design stage.

The 'Cost of Works' forecast

You will recall our discussion of the Cost of Works and that our fees, among other things, are not part of calculating the Cost of Works. It is very important that we proceed with a mutual understanding of what

the Cost of Works covers and how it is used during the stages of our services. We encourage you to read the enclosed 'Information guide to the client' document which also explains these concepts under the CAA.

Clause A.3 and Section M of the CAA (pages 2 and 7) set out the definition of the Cost of Works. If you are unclear on what parts of your Total Project Cost are included in the Cost of Works and which parts are not, please ask us or refer to the 'Information guide to the client' document.

On our current understanding of your design brief, and the quality and size of the project, our initial Cost of Works forecast is **$9,600,000** (excl. GST).

In deriving this initial Cost of Works forecast we have considered the cost of comparable large public buildings, but indicative costs are not a guarantee of actual costs nor the final cost for your project. The roof structure in particular has no precedent, and its cost cannot yet be estimated with confidence.

This preliminary indication of cost is provided to assist you in understanding your design brief and your **Cost of Works** budget as it relates to your **Total Project Cost**, discussed below. It is also based on assumptions we have made about the design brief and information available at this point in time.

As discussed, we highly recommend you engage a quantity surveyor or professional cost consultant to assist you in managing your overall budget for the project. A cost consultant can provide professional and more accurate estimates of the project costs, as the project takes shape.

Your budget and the Total Project Cost

The Total Project Cost (also defined in Section M) includes the Cost of Works plus all other costs that relate to the project including (but not limited to) our fees and those of other specialist consultants and professionals; disbursements; removal and relocation costs; design and construction contingencies including escalation costs; planning and other authority fees and charges; legal fees; landscaping; loose or unaffixed furniture, fittings and equipment; borrowing or finance costs; marketing and promotional costs, etc.

As discussed, you are responsible for managing the Total Project Cost budget, because this concept covers components we are unable to control, manage, or don't have responsibility for. Please also consider the GST components for costs, because the GST payable on the project costs (including our fees) should be factored in your budget.

We also recommend that you include in the Total Project Cost budget a contingency sum of say 30%, to allow for unforeseen expenses that may eventuate as the project proceeds. The roof has no precedent and the ground is untested, so we recommend a larger allowance than for an ordinary building. Part of that allowance should be held against variations: we will report to you in writing whenever the variations approved under the building contract add up to more than 5% of the contract sum, with their effect on the Cost of Works forecast (clause A.3.2).

A few examples of expenses which might arise that you do not presently expect could be:

- you might later decide to have us carry out extra services you have not presently anticipated;
- the services of another specialist consultant may become desirable or necessary;
- during the construction of the project, you may decide you want to change something which involves extra cost to you (variations); or
- unexpected events occur which cause delays to developing the project or the construction works.

We understand that more detailed consideration of your **Total Project Cost** may require adjustments to the funds available for the **Cost of Works**.

Fees for Architectural Services

Fees for Core Architectural Services:

The scope of these Services is listed in Schedule A: Section A of the draft CAA (page 9). Please note that specific responsibilities and services not included in our professional services are listed in clause A.2 (page 2).

We propose a percentage Fee of 6% of the Cost of Works (excluding GST as outlined above), for which a preliminary estimate is:

6% of $9,600,000 (forecast Cost of Works excluding GST) = $576,000 (excluding GST)

The breakdown of these fees across the stages of the Core Architectural Services is set out in Schedule B: Item 1 of the draft CAA (page 13), as follows. Our fee includes a Mobilisation Fee of **$10,000** (excluding GST), an advance payment on the fees payable for the services (see Section M), payable within 5 business days after we sign the agreement. It matches the prize money for the competition, and is deducted from the A.1 Concept design fee below.

Schedule B item	Service	Share of fee	% of Cost of Works / estimate (excl. GST)
A.1	Concept design	15%	0.90% = $86,400 (incl. $10,000 Mobilisation Fee)
A.2	Design development	15%	0.90% = $86,400
A.3	Planning/Development Application	Excluded	Crown project: approvals by the Government
A.4	Construction documentation	35%	2.10% = $201,600
A.5	Contractor selection	5%	0.30% = $28,800
A.6	Contract administration	30%	1.80% = $172,800
Total	Core Architectural Services	100%	6.00% = $576,000

These Core Services cover the whole path from the competition scheme to a completed building: developing the scheme with you, documenting it for tender, helping you select the building contractor, and administering the building contract on your behalf (ABIC Simple Works 2018), including variations, adjustments of time, progress certificates and the final certificate. We include contract administration because the building is technically unprecedented, and decisions on site will need the designer's judgement.

Fees for Additional Architectural Services:

As discussed with you, we have identified that the following architectural services are required in addition to the Core Architectural Services. These services are shown in Schedule A: Section B of the CAA (page 11) and will attract the fees set out in Schedule B: Item 1 for components B.1 to B.6 (page 13).

Individual services in Schedule A which are priced by a different fee method, for the purposes of Schedule B: Item 1, are set out below:

Schedule B item:	Brief description of service:	Fee method	Notes
B.4	Illustration: presentation drawings, models and a printed report of the developed scheme, for your Government and the public	Time-based rates (Item 2)	We will give you an estimate before starting

When considering your **Total Project Cost** budget you will need to consider our fees for both the Core Architectural Services and Additional Architectural Services.

Specialist Consultants

In our discussions with you, we have identified the need for specialist consultants to assist in the delivery of the project and these are listed in Clause E of the Client Architect Agreement.

Item 4 of Schedule B indicates the specialist consultants whose services we recommend are likely to be needed during the project. As discussed, these specialist consultants are to be engaged by you on the terms outlined in clause E.1, and their fees will need to be considered in your Total Project Cost budget. We will coordinate their work, but we are not responsible for the services they deliver (clause E.1.3).

We recommend:

- **Structural engineer, Ove Arup & Partners (London):** the roof shells and the podium that carries them are the central engineering problem of the project;
- **Mechanical and electrical engineer, Steensen & Varming (Copenhagen):** air conditioning, electrical services and lighting for two large halls;
- **Quantity surveyor,** Rider Hunt & Partners**:** cost planning at each stage and the bills of quantities for tender;
- **Geotechnical engineer:** to investigate the ground at Bennelong Point before the foundations are designed; and
- **Acoustic and stage-machinery consultants:** the halls must work for both opera and concerts.

We will make recommendations if the need for additional specialist consultants arises during the course of the project, and if they are engaged, clause E.2 will govern the arrangements.

Disbursements

We have discussed with you the items for which you would reimburse us their cost and those which would also attract a service fee. Those we have presently identified are set out in Item 3 of Schedule B of the Client Architect Agreement.

For those disbursements listed with a service fee applicable, clause D.7 of the Client Architect Agreement provides that the service fee payable is 15 per cent of the invoiced cost of a disbursement.

Because our office is in Hellebæk, Denmark, the main disbursements will be overseas travel between Denmark and Sydney, international courier and postage, and presentation models. If these are likely to be significant, you may wish to consider their likely total amount in your Total Project Cost budget.

Program of services

You have told us that you want to start building as soon as possible. The roof structure is not yet resolved, and its loads will govern the design of the podium and its columns. We therefore set out two ways of running the project, so that you can choose with the risk in front of you. We have prepared the draft CAA on Option 1 because it is your stated preference.

Option 1: staged (fast-track). The work is divided into three stages (podium; roof shells; interiors), each with its own design, documentation and building contract. The podium is designed and documented in about 18 to 24 months, with a presentation of the developed scheme to you in early 1958, and building starts while the roof is still being designed. The podium takes about 4 years to build. **Risk:** the roof's real loads may exceed those assumed for the podium, so columns already built may have to be strengthened or rebuilt. That would be a variation to the building contract and a change to our services (clause D.8), with extra time and cost, and the later stages each repeat the tendering process.

Option 2: sequential. The roof structure and the whole building are resolved with the engineers, documented and tendered before building starts. We estimate 5 to 6 years of design and documentation, then 6 to 8 years of construction. **Benefit:** the podium is built once, to the final loads, and the cost forecast is much firmer when you commit. **Cost:** the building opens later, and your Government waits longer to see work on site.

Both options depend on factors we cannot forecast: the time to obtain the necessary Government approvals, and the time to resolve the roof structure with the engineers. If you choose Option 1, we recommend you record the risk above in the program and in the contingency, and we will tell you promptly, in writing, if the design shows that built work must change.

In addition, contributing to the indicated timeframe, there are the necessary approvals for building work to commence and the selection of the contractor. Similarly, we are not in a position to forecast the time involved before the construction work can begin.

Generally, after the building contractor is selected, a building contract signed, and approvals and finance are secured, construction work can start.

Communication, changes and variations

Our representative will be Jørn Utzon. Please confirm your Government's representative for the CAA (clause C.2). If either representative changes, the other party must be told in writing (clause C.3).

All notices, approvals and instructions will be in writing, and email counts (clause H.1). Because our office is overseas, a posted notice is deemed received 10 business days after posting (clause H.8), so we will also send every notice by email.

We will report progress at each stage, and ask for your written approval before moving to the next stage (Schedule A).

If your instructions or the project's needs change the scope of our services, we will promptly tell you in writing how this affects our fee, and agree the change with you in writing before we carry it out (clauses A.1.4 and D.8).

Once the building contract is signed, instructions to the builder must be issued only through us (clause B.2). Changes to the building work will be issued as variations under the building contract, with their effect on time and cost assessed and recorded.

Engaging us to work further with you

We confirm our enthusiasm for providing architectural services for your project and look forward to exploring the opportunities that your project offers.

However, before we do further work we request that you formally confirm our engagement, in writing.

Under the Architects Act and the Code of Conduct that applies to our professional services, our engagement with you must be recorded and confirmed in a written agreement before we start delivering professional services to you. As discussed and noted above, we propose our engagement is formalised by signing the Australian Institute of Architects' Client Architect Agreement 2024 that is enclosed with this letter in draft form. We encourage you to read it carefully, seek legal advice if you wish and discuss with us any queries you have.

We again encourage you to read the 'Information guide to the client' document to assist you in understanding the CAA and other important elements of our engagement.

We look forward to your confirmation. We will then prepare final copies of the CAA and send them to you to sign and return.

Please do not hesitate to call us if you have any questions.

Yours sincerely,

Jørn Utzon, Architect, Hellebæk, Denmark

Jørn Utzon
Principal